Fernando Contreras
Assignment 9.2: APIs
May 6, 2026

Part 1 — Connection Test (api_test_fc.py)

api_test_fc.py 1, U

csd-325 > module-9 > api_test_fc.py > ...

```python
"""
api_test_fc.py
Author: Fernando Contreras
Assignment: CSD-325 Module 9
Description: Tests connection to a URL using the requests library.
             Part 1 of the API tutorial — verifying the requests module works.
"""

import requests

# Test connection to Open Notify API
url = 'http://api.open-notify.org/astros.json'
response = requests.get(url)
print(response.status_code)

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh - module-9

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-9 % ls
api_astronauts_fc.py    api_main_fc.py          api_test_fc.py
(base) fernandocontreras@Fernandos-MacBook-Pro module-9 % python api_test_fc.py
200
(base) fernandocontreras@Fernandos-MacBook-Pro module-9 % python api_test_fc.py
200
(base) fernandocontreras@Fernandos-MacBook-Pro module-9 %
```

Part 2 — Astronaut Tutorial (api_astronauts_fc.py)

api_test_fc.py 1, U | api_main_fc.py 1, U | api_astronauts_fc.py 1, U

csd-325 > module-9 > api_astronauts_fc.py > ...

```
"""
api_astronauts_fc.py
Author: Fernando Contreras
Assignment: CSD-325 Module 9
```

PROBLEMS 3 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | zsh -

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-9 % python api_astronauts
--- Connection Test ---
Status code: 200

--- Current Astronauts in Space ---
Number of people in space: 12

Name: Oleg Kononenko
Craft: ISS
---
Name: Nikolai Chub
Craft: ISS
---
Name: Tracy Caldwell Dyson
Craft: ISS
---
Name: Matthew Dominick
Craft: ISS
---
Name: Michael Barratt
Craft: ISS
---
Name: Jeanette Epps
Craft: ISS
---
Name: Alexander Grebenkin
Craft: ISS
---
Name: Butch Wilmore
Craft: ISS
---
Name: Sunita Williams
Craft: ISS
---
Name: Li Guangsu
Craft: Tiangong
---
Name: Li Cong
Craft: Tiangong
---
Name: Ye Guangfu
Craft: Tiangong
---
(base) fernandocontreras@Fernandos-MacBook-Pro module-9 %
```

Part 3 — Custom API Program (api_main_fc.py)

API chosen: JokeAPI (https://v2.jokeapi.dev) — a free API that returns programming jokes in JSON format. The program tests the connection, prints the raw response, then prints a formatted version of the data following the same pattern as the tutorial.

api_test_fc.py 1, U api_main_fc.py 1, U

csd-325 > module-9 > api_main_fc.py > ...

```python
"""
api_main_fc.py
Author: Fernando Contreras
Assignment: CSD-325 Module 9
Description: Uses the JokeAPI (https://v2.jokeapi.dev) to retrieve joke
             Tests the connection, prints the raw response, then prints

API chosen: JokeAPI — free, returns JSON data.
URL: https://v2.jokeapi.dev/joke/Programming?amount=3
"""

import requests
import json

API_URL = 'https://v2.jokeapi.dev/joke/Programming?amount=3'

# — Step 1: Test the connection ————————————————————————
print("--- Connection Test ---")
response = requests.get(API_URL)
print(f"Status code: {response.status_code}")
print()

# — Step 2: Print raw (unformatted) response ————————————
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS zsh - m

```
(base) fernandocontreras@Fernandos-MacBook-Pro module-9 % python api_main_fc.py
--- Connection Test ---
Status code: 200

--- Raw Response ---
{
    "error": false,
    "amount": 3,
    "jokes": [
        {
            "category": "Programming",
            "type": "twopart",
            "setup": "What's the object-oriented way to become wealthy?",
            "delivery": "Inheritance.",
            "flags": {
                "nsfw": false,
                "religious": false,
                "political": false,
                "racist": false,
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS


```
                "religious": false,
                "political": false,
                "racist": false,
                "sexist": false,
                "explicit": false
            },
            "id": 17,
            "safe": true,
            "lang": "en"
        },
        {
            "category": "Programming",
            "type": "twopart",
            "setup": "Hey, wanna hear a joke?",
            "delivery": "Parsing HTML with regex.",
            "flags": {
                "nsfw": false,
                "religious": false,
                "political": false,
                "racist": false,
                "sexist": false,
                "explicit": false
            },
            "id": 10,
            "safe": true,
            "lang": "en"
        }
    ]
}
```


```
--- Formatted Response ---
Number of jokes retrieved: 3

Joke 1:
  Category : Programming
  Setup    : What's the object-oriented way to become wealthy?
  Delivery : Inheritance.

Joke 2:
  Category : Programming
  Setup    : How can you tell an extroverted programmer?
  Delivery : He looks at YOUR shoes when he's talking.

Joke 3:
  Category : Programming
  Setup    : Hey, wanna hear a joke?
  Delivery : Parsing HTML with regex.

(base) fernandocontreras@Fernandos-MacBook-Pro module-9 %
```